DE–3653

Sub. Code
11

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Suppose that H is a subgroup of G such that whenever $H_a \neq H_b$ then $aH \neq bH$. Prove that $gHg^{-1} \subset H$, for all $g \in G$.

   (b) Prove that N is a normal subgroup of G if and only if $gNg^{-1} = N$ for every $g \in G$.

2. (a) Show that if $p^n \mid O(G)$, $p^{n+1} \nmid 0(G)$ then G has a subgroup of order p^n.

   (b) If G is the internal direct product of $N_1, N_2, \cdots, N_n$ prove that for $i \neq j$, $N_i \cap N_j = \{e\}$, and if $a \in N_i$, $b \in N_j$ then $ab = ba$.

3. (a) Show that every Euclidean ring is a principal ideal ring.

   (b) Prove : If R is a commutative ring with unit element and M is an ideal of R, then M is a maximal ideal of R if and only if $\frac{R}{M}$ is a field.

4. (a) Prove that every integral domain can be imbedded in a field.

   (b) State and prove the unique factorization theorem.

5. (a) Prove that if V is finite-dimensional vector space over F then V is isomorphic to $F^{(n)}$ for a unique integer n, in fact, n is the number of elements in any basis of V over F.

   (b) Prove that if V is a finite dimensional vector space, then it contains a finite set $v_1, v_2 \cdots v_n$ of linearly independent elements whose linear span is V.

6. (a) If S and T are nilpotent linear transformations which commute, prove that ST and $S+T$ are nilpotent linear transformations.

   (b) If $T \in A(V)$ is Hermitian, then all its characteristics roots are real.

7. (a) Prove that if L is a finite extension of F and K is a subfield of L which contains F, then $[K:F]/[L:F]$.

   (b) Prove that the polynomial $f(x) \in F[x]$ has δ multiple roots if and only if $f(x)$ and $f'(x)$ have a non trivial common factor.

8. (a) If p is a prime number, prove that the spliting field over F, the field of rational roots of the polynomial $x^p - 1$ is of degree $p-1$.

   (b) If the finite field F has p^m elements, then prove the every $a \in F$ satisfies $a^{p^m} = a$.

DE–3654

Sub. Code
12

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

REAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

($5 \times 20 = 100$)

1. (a) State and prove representation theorem for open sets on the real line. (10)

   (b) State and prove Bolzano-Weierstrass theorem. (10)

2. (a) Prove that in an Euclidean space R^K, every Cauchy sequence is convergent. (10)

   (b) State and prove Heine theorem. (10)

3. (a) State and prove chain rule. (10)

   (b) State and prove generalized mean-value theorem. (10)

4. State and prove implicit function theorem. (20)

5. (a) If P^* is a refinement of P, then prove that $L(p,f,\alpha) \le (P^*,f,\alpha)$ and $U(P^*,f,\alpha) \le U(P,f,\alpha)$. (10)

   (b) Assume that α increases monotonically and $\alpha' \in \mathcal{R}$ on $[a,b]$. Let f be a bounded real function on $[a,b]$, then prove that $f \in \mathcal{R}(\alpha)$ if and only if $f\alpha' \in \mathcal{R}$. In that case $\int_a^b f d\alpha = \int_a^b$. (10)

6. (a) Prove that the outer measure of an interval is equal to its length. (10)

(b) State and prove Lebesgue monotone convergence theorem. (10)

7. (a) Suppose $f_n \to f$ uniformly on a set E in a metric space. Let x be a limit point of E and suppose that $\lim_{t \to x} f_n(t) = A_n \; (n = 1,2,3,...)$ then prove that $\{A_n\}$ converges and $\lim_{t \to x} f(t) = \lim_{n \to \infty} A_n$. (10)

(b) Suppose K is compact and if (10)

(i) $\{f_n\}$ is a sequence of continuous functions on K,

(ii) $\{f_n\}$ converges point wise to a continuous function f on K,

(iii) $f_n(x) \geq f_{n+1}(x) \forall x \in K, n = 1,2,...$ then prove that $f_n \to f$ uniformly on K.

8. (a) State and prove Egoroff's theorem. (10)

(b) State and prove Fatou's Lemma. (10)

DE–3655

Sub. Code
13

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Find the solutions of the following initial value problem :

(i) $y''-2y'-3y=0, y(0)=0, y'(0)=1$

(ii) $y''+(3i-1)y'-3iy=0, y(0)=2, y'(0)=0$ (10)

(b) Find all the solutions of the equation

$x^2y''-3xy^1-5y=0$ for $|x|>0$. (10)

2. (a) Let ϕ_1, ϕ_2 be two solutions of $L(y)=0$ on an interval I, and let x_0 be any point in I. Prove that ϕ_1, ϕ_2 are linearly independent on I if and only if $W(\phi_1, \phi_2)(x_0) \neq 0$. (10)

(b) Find the power series solution of the Legendre equation $(1-x^2)y''-2xy'+\alpha(\alpha+1)y=0$. (10)

3. (a) Show that the equations $xp - yq = x$, $x^2p + q = xz$ are compatible and solve them. (10)

(b) Show that the only integral surface of the equation $2q(z - px - qy) = 1 + q^2$ which is circumscribed about the paraboloid $2x = y^2 + z^2$ is the enveloping cylinder which touches it along its section by the plane $y + 1 = 0$. (10)

4. (a) Solve the equation $q^2r - 2pqs + p^2t = 0$. (10)

(b) Reduce the equation $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ to canonical form and hence solve it. (10)

5. (a) Solve $x \tan x = -1$ by Newtons method. (10)

(b) Solve $e^x - 3x^2 = 0$ by fixed point method. (10)

6. (a) Solve

$$\begin{aligned} 20x + y - 2z &= 17 \\ 3x + 20y - z &= -18 \\ 2x - 3y + 20z &= 25 \end{aligned}$$

by iteration method. (10)

(b) Solve the relaxation method.

$$\begin{aligned} 5x_1 - x_2 - x_3 &= 3 \\ -x_1 + 10x_2 - 2x_3 &= 7 \\ -x_1 - x_2 + 10x_3 &= 8. \end{aligned}$$ (10)

7. Find the first, second and third derivatives of the following function at the point $x = 1.5$. (20)

x	1.5	2.0	2.5	3.0	3.5	4.0
$f(x)$	3.375	7.0	13.625	24.0	38.875	59.0

8. Solve the equation $\frac{d^2y}{dx^2} = xy^2$, $y(0) = 1$, $y'(0) = 0$ for $y(0.2)$ and $y(0.4)$ by $R-K$ method of fourth order. (20)

DE–3656

Sub. Code
14

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

Each question carries equal marks.

1. (a) Write the procedure for simplex method by solving L.P.P. (8)

(b) Solve the L.P.P. by revised simplex method. (12)

Minimize $z = 5x_1 - 4x_2 + 6x_3 + 8x_4$

subject to

$$x_1 + 7x_2 + 3x_3 + 7x_4 \le 46$$
$$3x_1 - x_2 + x_3 + 2x_4 \le 20$$
$$2x_1 + 3x_2 - x_3 + x_4 \ge 18 \text{ and}$$
$$x_1, x_2, x_3, x_4 \ge 0$$

2. (a) Solve the L.P.P by revised dual simplex method. (10)

Maximize $z = 2x_1 + x_2 + 2x_3$

subject to

$$4x_1 + 3x_2 + 8x_3 \le 12$$
$$4x_1 + x_2 + 12x_3 \le 8$$
$$4x_1 - x_2 + 3x_3 \le 8 \text{ and}$$
$$x_1, x_2, x_3 \ge 0$$

(b) Solve the following parametric L.P.P. (10)

Maximize $z = 3x_1 + 2x_2 + 5x_3$

subject to

$$x_1 + 2x_2 + x_3 \le 40 - t$$
$$3x_1 + 2x_3 \le 60 + 2t$$
$$x_1 + 4x_2 \le 30 - 7t \quad \text{and}$$
$$x_1, x_2, x_3 \ge 0$$

3. (a) Solve the following problem by dynamic programming. (10)

Maximize $z = 4x_1 + 14x_2$

subject to

$$2x_1 + 7x_2 \le 21$$
$$7x_1 + 2x_2 \le 21 \text{ and } x_1, x_2 \ge 0.$$

(b) State the 'Principle of optimality' in dynamic programming and give a mathematical formulation of a dynamic programming problem. (10)

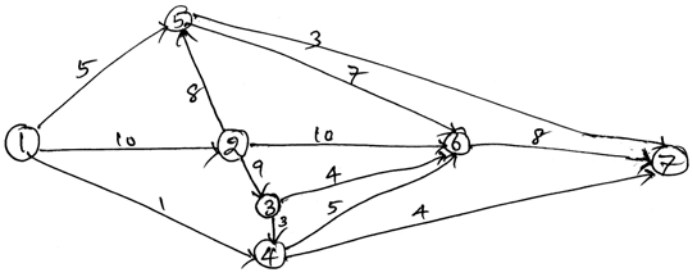
4. (a) Solve the following 2×2 game graphically : (10)

	B_1	B_2	B_3	B_4
A_1	2	2	3	–1
A_2	4	3	2	6

(b) Solve the following game by linear programming. (10)

	B_1	B_2	B_3
A_1	3	–1	–3
A_2	–2	4	–1
A_3	–5	–6	2

5. (a) Determine the critical path for the project network. (10)



(b) (i) What is the difference between PERT and CPM? (5 + 5 = 10)

(ii) Describe the total float and free float of an activity.

6. (a) Lube car specializes in fast automobile oil change. The garage buys car oil in bulk at \$ 3 per gallon. A price discount of \$ 2.50 per gallon is available if Lube car purchases more than 1000 gallons. The garage services approximately 150 cars per day, and each oil change requires 1.25 gallons. Lube car stores bulk oil at the cost of \$ 0.02 per gallon per day. Also the cost of placing an order for bulk oil is \$ 20. There is a 2-day lead time for delivery. Determine the optimal inventory policy. (10)

(b) Enumerate the no-setup model. (10)

7. (a) Babies are born in a sparsely populated state at the rate of one birth every 12 minutes. The time between births follows an exponential distribution. Determine the following : (10)

(i) The average number of births per year.

(ii) The probability that no births will occur in any one day.

(iii) The probability of issuing 50 birth certificates by the end of the next 3 hours given that 40 certificates were issued during the last 2 hours.

(b) Explain the pure birth and pure death models in queueing theory. (10)

8. (a) Visitors parking at Ozark college is limited to only five spaces. Cars making use of this space arrive according to a Poisson distribution at the rate of six cars per hour. Parking time is exponentially distributed with a mean of 30 minutes. Visitors who cannot find an empty space immediately on arrival may temporarily wait inside the lot until a parked car leaves. That temporary space can hold only three cars. All other cars that cannot park or find a temporary waiting space must go elsewhere. Determine the following : (10)

(i) The probability p_n of n cars being in the system.

(ii) The effective rate at which cars arrive at the lot.

(iii) The average number of cars in the lot.

(iv) The average time a car waits for a parking space inside the lot.

(b) Derive L_q, L_s, W_s, W_q for $(M/M/1) : (GD/N/\infty)$ queueing model. (10)

DE–3657

Sub. Code
15

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

MATHEMATICAL STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) A continuous random variable has a pdf $f(x) = kx^2 e^{-x}$, $x > 0$. Find k, mean and variance. (10)

(b) The joint density function of the random variables X and Y is given by

$$f(x, y) = \begin{cases} 8xy, & 0 < x < 1;\ 0 < y < x \\ 0, & \text{elsewhere}. \end{cases}$$

Find $P\left(Y < \frac{1}{8} \Big/ X < \frac{1}{2}\right)$. (10)

2. (a) Verify central limit theorem for the independent random variables X_k, where for each k, $p(X_k = \pm 1) = \frac{1}{2}$. (10)

(b) If X and Y are independent random variables following $N(8, 2)$ and $N(12, 4\sqrt{3})$ respectively, find the value of λ such that $p(2X - Y \le 2\lambda) = P(X + 2Y \ge \lambda)$ (10)

3. (a) Discuss about constants of t-distribution. (10)

(b) If $X_1, X_2, ..., X_n$ are Poisson variates with parameter $\lambda = 2$, use central limit theorem to estimate $P(120 \le S_n \le 160)$, where $S_n = X_1 + X_2 + ... + X_n$ and $n = 75$. (10)

4. (a) If T_1 and T_2 are two unbiased estimator of $\gamma(\theta)$, having the same variance and ρ is the correlation between them, then show that $\rho \ge 2e - 1$, where e is the efficiency of each estimator. (10)

(b) Discuss the maximum likelihood estimation. (10)

5. (a) Give the major steps involved in the solution of a testing of hypothesis problem. (10)

(b) Let p be the probability that a coin will fall head in a single toss in order to test $H_0 : p = \frac{1}{2}$ against $H_1 : p = \frac{3}{4}$. The coin is tossed 5 times and H_0 is rejected, if more than 3 heads are obtained. Find the probability of type 1 error and power of the test. (10)

6. (a) Discuss about moments of multinomial distribution. (10)

(b) If $X_1 + X_2 + ... + X_k$ are k independent Poisson variates with parameters $\lambda_1, \lambda_2, ..., \lambda_k$ respectively, prove that the conditional distribution :

$P(X_1 \cap X_2 \cap ... \cap X_k / X)$,

where $X = X_1 + X_2 + ... + X_k$ is fixed, is multinomial. (10)

7. (a) Explain the control charts : np-chart, p-chart and c-chart. (10)

   (b) Discuss about control chart for sample standard deviation. (10)

8. (a) Explain the analysis of variance for one factor of classification. (10)

   (b) The varieties of a crop are tested in a randomised block design with four replications, the layout being as given below. The yields are given in kilograms. Analyse for significance. (10)

C48	A51	B52	A49
A47	B49	C52	C51
B49	C53	A49	B50

DE–3658

Sub. Code
21

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) (i) Is an analytic function $f(2)$ with a constant modulus reduces to a constant? If yes, justify your answer. (5)

(ii) If $u = x^2 - y^2$ is harmonic, find v such that $u + iv$ is analytic. (5)

(b) Define cross ratio. If z_1, z_2, z_3, z_4 are distinct points in the extended plane and T any linear transformation, then prove that $(Tz_1, Tz_2, Tz_3, Tz_4,) = (z_1, z_2, z_3, z_4)$. (10)

2. (a) State and prove Abel's limit theorem. (12)

(b) Discuss the transformation $w = \sin z$. (8)

3. (a) State and prove Cauchy's theorem for a rectangle. (10)

(b) State and prove the local mapping theorem. (10)

4. (a) State and prove the Schwarz lemma. (10)

(b) State and prove Liouville's theorem. Deduce that the fundamental theorem of algebra. (10)

5. (a) Obtain the Laurent expansion $\sum_{n=-\infty}^{\infty} A_n (z-a)^n$ for the function $f(z)$ analytic in $R_1 < |z-a| < R_2$. (10)

(b) Evaluate $\int_0^{\infty} \frac{\cos x}{x^2 + a^2} dx$, (a real). (10)

6. (a) With the usual notations, prove that $\frac{\pi^2}{\sin^2 \pi z} = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}$. (10)

(b) Derive the Legendre's duplication formula. (10)

7. (a) Show that any two bases of the same module, are connected by a unimodular transformation. (10)

(b) Prove that the zeros $a_1, a_2, \ldots. a_n$ and poles $b_1, b_2, \ldots. b_n$ of an elliptic function satisfy $a_1 + a_2 + \ldots. + a_n \equiv b_1 + b_2 + \ldots + b_n$ (mod M). (10)

8. (a) Show that $p'(z) = -2 \sum_{w} \frac{1}{(z-w)^3}$. (10)

(b) With the usual notations, prove that $p(2z) = \frac{1}{4}\left(\frac{p''(z)}{p'(z)}\right)^2 - 2p(z)$. (10)

DE–3659

Sub. Code
22

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

TOPOLOGY AND FUNCTIONAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Let X be a topological space. Then prove that any intersection of closed sets in X is closed and any finite union of closed sets in X is closed. (10)

   (b) Let $f : X \to Y$ be a mapping of one topological space into another. (10)

   Prove that the following are equivalent :

   (i) f is continuous

   (ii) $f^{-1}(F)$ is closed in X, whenever F is closed in Y.

   (iii) $f(\overline{A}) \subseteq \overline{f(A)}$ for every subset A of X.

2. (a) Show that a topological space is compact iff every class of closed sets with empty intersection has a finite subclass with empty intersection. (7)

   (b) Prove that a topological space is compact if every basic open cover has a finite sub cover. (7)

   (c) Show that a metric space is sequentially compact iff it has the Bolzano Weierstrass property. (6)

3. (a) State and prove the *Urysohn Imbedding theorem*. (10)

   (b) Show that the product of any one-empty class of connected spaces is connected. (10)

4. (a) Prove that the spaces R^n and C^n are connected. (12)

   (b) Prove that every compact subspace of a Hausdorff space is closed. (8)

5. (a) Let M be a closed linear subspace of a Banach space N. If the norm of a coset $x+M$ in N/M is defined by $\|x+M\| = \inf\{\|x+m\| : m \in M\}$, then show that N/M is a Banach space. (10)

   (b) If B and B' are Banach spaces and if T is a continuous linear transformation of B onto B', then prove that T is an open mapping. (10)

6. (a) State and prove *closed graph theorem*. (10)

   (b) State and prove the 'Uniform Boundedness theorem'. (10)

7. (a) State and prove 'Schwartz inequality'. (8)

   (b) If M is a proper closed linear subspace of a Hilbert space H. Then show that there exist a vector z_0 in H such that $z_0 \perp M$. (12)

8. (a) Let H be a Hilbert space, and let f be an arbitrary functional in H^*. Prove that there exists unique vector y in H such that $f(x) = (x, y)$ for every x in H. (10)

   (b) Prove that an operator T on H is self-adjoint if and only if (Tx, x) is real for all x. (10)

DE–3660

Sub. Code
23

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

GRAPH THEORY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that a graph bipartite if and only if it contains no odd cycle. (10)

   (b) If e is a link of G, prove that $Z(G) = Z(G - e) + Z(G.e)$. (10)

2. (a) Establish Cayley's formula for number of spanning trees. (10)

   (b) State and prove Sperner's lemma. (10)

3. (a) Define vertex connectivity and edge connectivity of a graph . (4)

   (b) Prove that for any graph , the vertex connectivity the edge connectivity the minimum degree. (8)

   (c) Prove that $C(G)$, the closure of G is well defined. (8)

4. (a) Prove that the graph $H_{m,n}$ is m-connected. (10)

   (b) State and prove Whitney theorem. (10)

5. (a) For any two integers $k \geq 2$ and $l \geq 2$, prove that $r(k,l) \leq r(k,l-1) + r(k-1,l)$ Also prove that, if $r(k,1-1)$ and $r(k-1,l)$ are both even then $r(k,l) < r(k,l-1) + r(k-1,l)$. (12)

(b) State and prove Brook's Theorem. (8)

6. (a) Prove that every planar graph is 5-vertex colourable. (10)

(b) Define *Hajo's conjecture* and hence show that if G is 4-Chromatic then G contains a subdivision of K_4. (10)

7. (a) Let G be a non planar connected graph that contains no subdivision of K_5 or $K_{3,3}$ and few edges as possible. Then prove that G is simple and 3–connected. (10)

(b) If G is planar, prove that every sub graph of G is planar. (10)

8. (a) Prove that the value of any flow in a network N is less than or equal to the capacity of any cut in the network. (10)

(b) State and prove max-flow, min-cut theorem. (10)

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DE–3661

Sub. Code
24

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2016.

PROGRAMMING IN C/C++

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) Explain constants, variables and their declaration in 'C'. (10)

   (b) Explain library functions used in C language in details with examples. (10)

2. (a) Discuss the structure of a C program with example. (10)

   (b) Explain the concept of object oriented programming. (10)

3. (a) Explain puts() and putchar() functions with examples. (10)

   (b) What do you mean by control codes and arithmetic operators? (10)

4. (a) Discuss the arithmetic operators in C programming with examples. (10)

   (b) Write short notes on :

      (i) Counters

      (ii) Increment and

      (iii) Decrement operators with examples. (10)

5. (a) Explain for loop with suitable program in C++. (10)

   (b) Explain Nested if statements with an example. (10)

6. (a) Define arrays and initialization of arrays in details with examples. (10)

   (b) Explain :

      (i) Assigning and combining strings (5)

      (ii) String arrays. (5)

7. (a) Discuss about pointers and functions. (10)

   (b) Explain structure and functions with suitable examples. (10)

8. (a) Explain the commands used to open and close the file. Explain in detail with example. (10)

   (b) Write a program to copy the contents of one file into another. (10)

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DE–3662

Sub. Code
25

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2016.

DISCRETE AND COMBINATORIAL MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) Determine the generating function for the numeric function given by $a_r = \begin{cases} 2^r & \text{if } r \text{ is even} \\ 2^{-r} & \text{if } r \text{ is odd.} \end{cases}$ (10)

   (b) Solve : $a_n - 4a_{n-1} + 4a_{n-2} = (n+1)^2$ given $a_0 = 0$ and $a_1 = 1$. (10)

2. (a) Solve : $a_n - 2a_{n-1} = (n+1)2^n$. (10)

   (b) In how many ways 12 of the 14 people will be distributed into 3 teams of 4 each? (10)

3. (a) Determine the number of integers between 1 and 250 that are divisible by any of the integers 2, 3, 5 and 7. (10)

   (b) Verify the principle of inclusion and exclusion for

   $A = \{x / x \text{ is a positive integer} < 8\}$

   $B = \{x / x \text{ is an integer and } 2 \le x < 4\}$

   $C = \{x / x \text{ is an integer and } x^2 < 16\}$. (10)

4. (a) Let $S = \{1,2,3,4\}$ and let $A = S \times S$. Define the relation R on $A : (a,b)\ R(a',b')$ if and only if $a + b = a' + b'$. Show that R is an equivalence relation and compute A / R. (10)

(b) State and prove Polya's fundamental theorem. (10)

5. (a) Show that the composition of permutation on S_3 is not an abelian group. (10)

(b) Let $A = \{a_1, a_2, ..., a_n\}$ be a finite set with n elements and $n > 2$. Prove that there are $\frac{n!}{2}$ even permutations and $\frac{n!}{2}$ odd permutations. (10)

6. (a) Explain the Pigeon hole principle with an example. (10)

(b) In a survey of 260 computer science students, 94 like to work in USA, 64 like to work in UK, 58 like to work in Singapore, 28 like UK and Singapore, 26 USA and UK, 22 USA and Singapore and 14 like all the three places. Find

(i) How many students like non of the three countries?

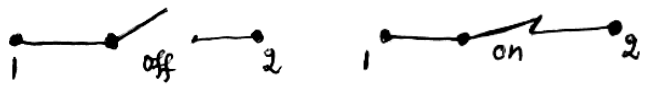
(ii) How many students like only USA? (10)

7. (a) Prove that in a lattice $(L, \leq)$, for any $a, b, c \in L$, if $a \leq b \leq c \Rightarrow a \vee b = bc$ and

$(a \wedge b) \vee (b \wedge c) = b = (a \vee b) \wedge (a \vee c)$. (10)

(b) In a distributive lattice, if $b \wedge \bar{c} = 0$, then prove that $b \leq c$. (10)

8. (a) Simplify the following function by Karnaugh map

$$f = x_1'x_2' + x_1x_3x_4 + x_1x_2x_4 + x_1x_2x_4' + x_2'x_3 .$$ (10)

(b) Explain the switching circuit of the following diagram of a simple on-off electric switch that can be placed either in the off position or in the on position. (10)